Homework

DM4 - A maximum and an equation - 60 min.

Consider the function f defined on $[0;+\infty[$ by $f(x) = (x+1)e^{-x}$. Its graph is denoted $\mathcal{C}_f$.

1. Calculate $f(0)$ and the limit of f in $+\infty$. Calculate $f'(x)$, then draw the variation table of f.

2. Calculate $f''(x)$. Study the convexity of f and determine the coordinates of the possible point of inflection of $\mathcal{C}_f$.

3. Prove that the equation $f(x) = \frac{1}{2}$ has a unique solution α on $[0;+\infty[$.

4. Write a Python function that uses bisection on $[0;2]$ and returns two real numbers a and b such that $a \leq \alpha \leq b$ and $b - a \leq 10^{-3}$. Justify the choice of the initial interval.

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. The position of the curve relative to relative to this single tangent enough to assert that f is concave on its whole domain? Justify.